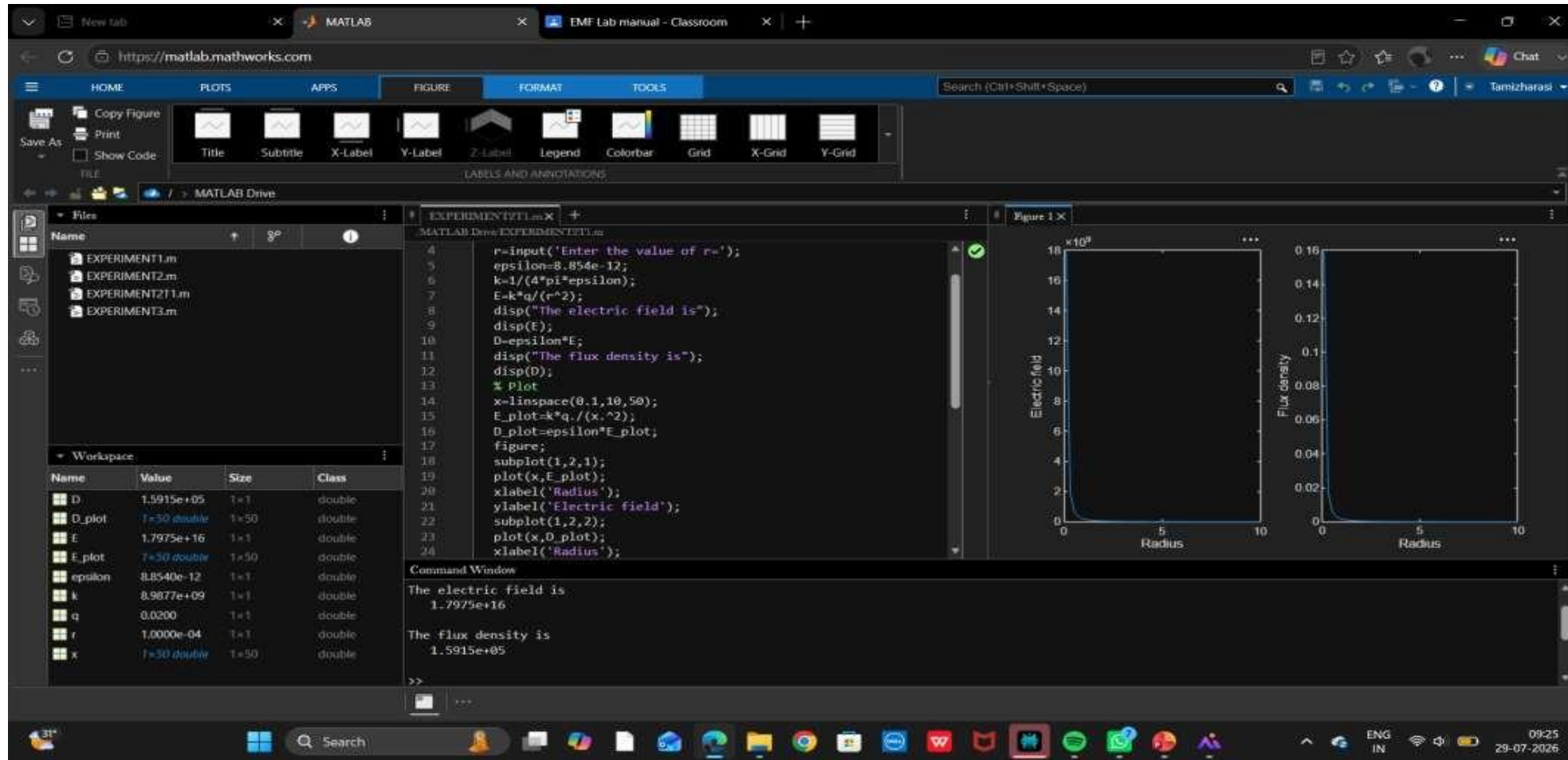
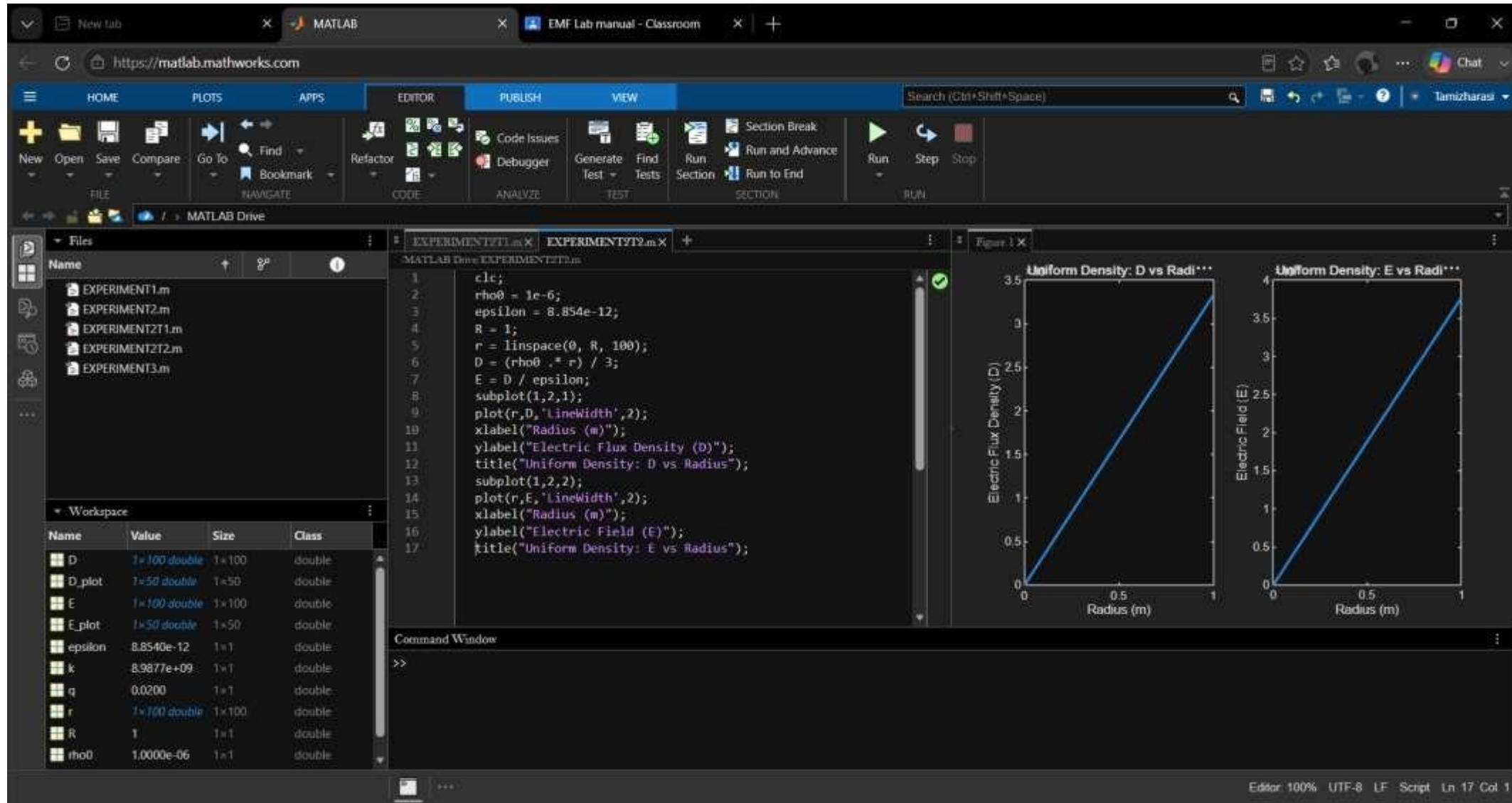


Exp. No: 2 Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

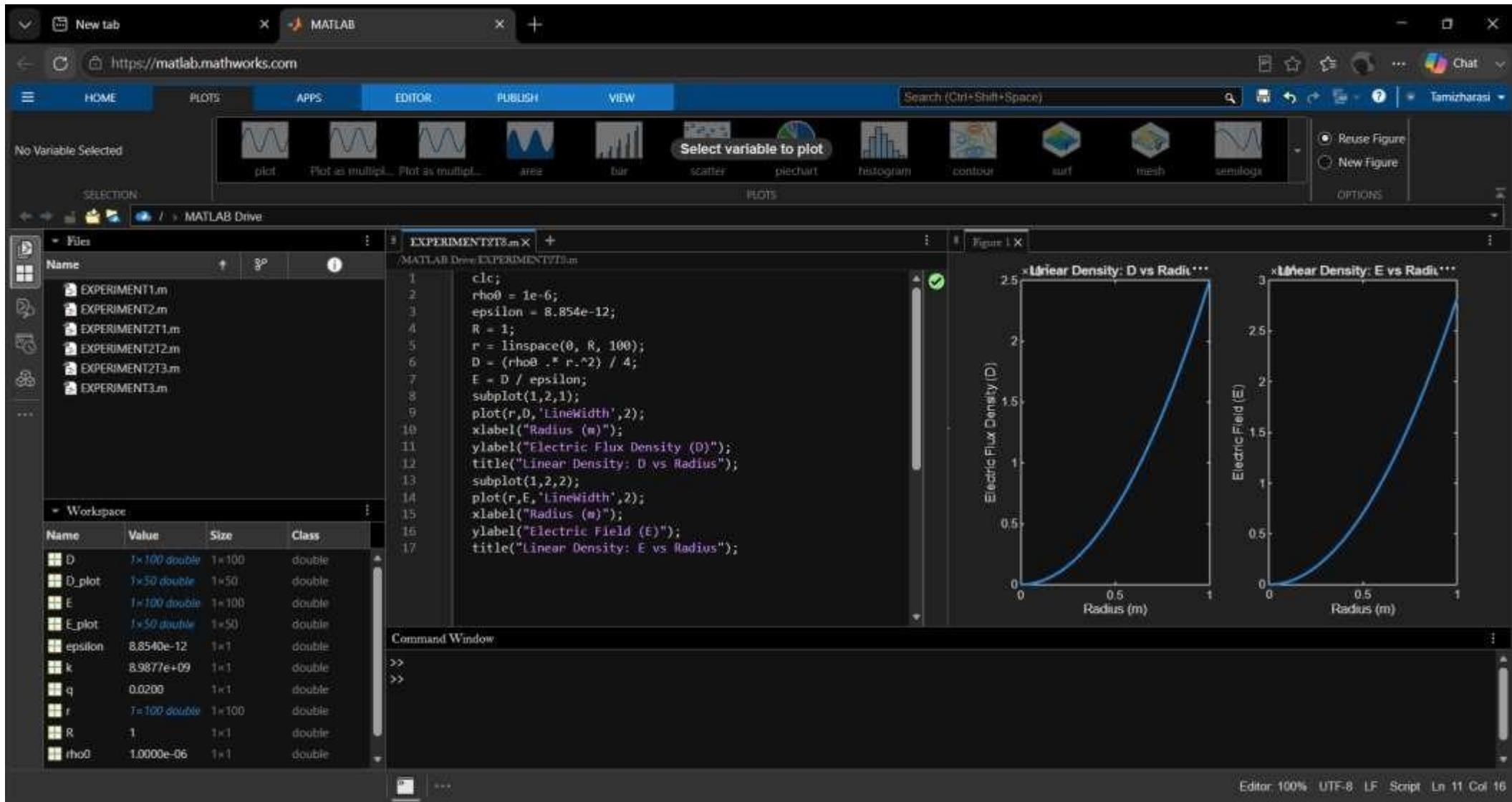
Test case (i) Determination and plotting of electric field and electric flux density in sphere full of charges with varying radius



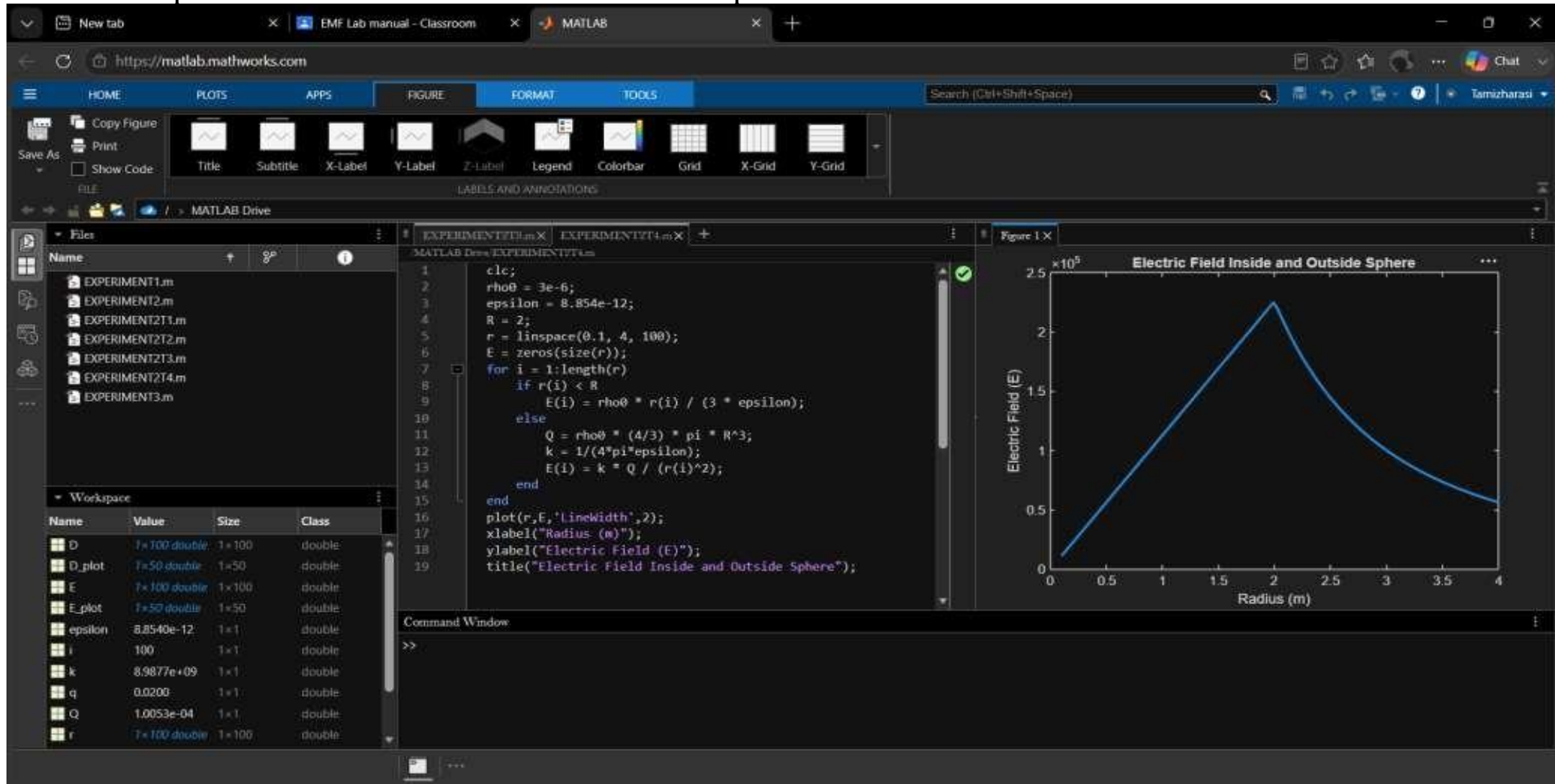
Test case (ii) Derive and plot the electric field (E) and electric flux density(D) inside a uniformly charged sphere with charge density ρ_0



Test case (iii) For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive and plot the electric field (E) and electric flux density (D)

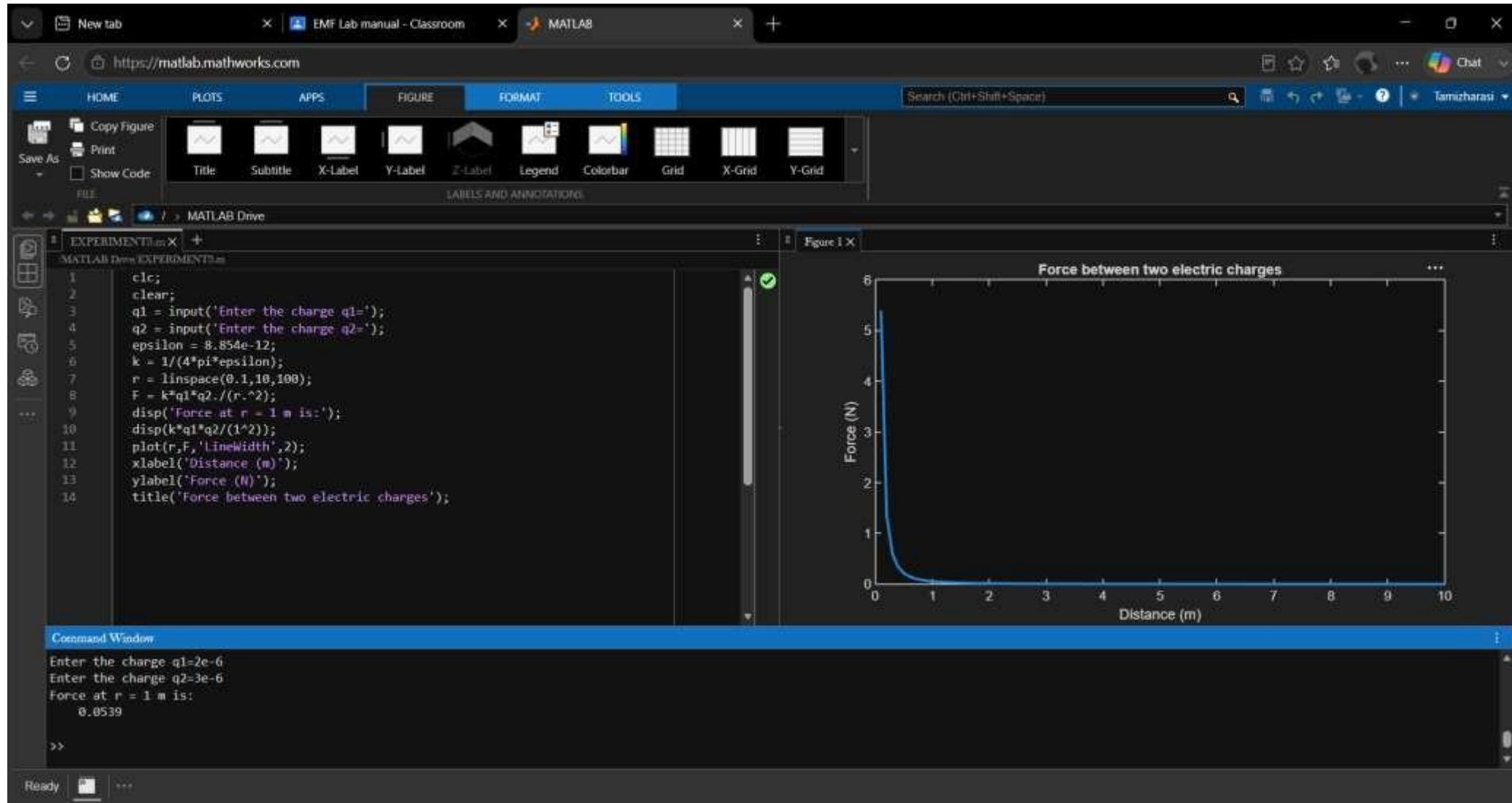


Test case (iv) A sphere of radius $R = 2$ m has a uniform charge density $\rho = 3 \times 10^{-6}$ C/m³. Calculate and plot electric field inside and outside the sphere

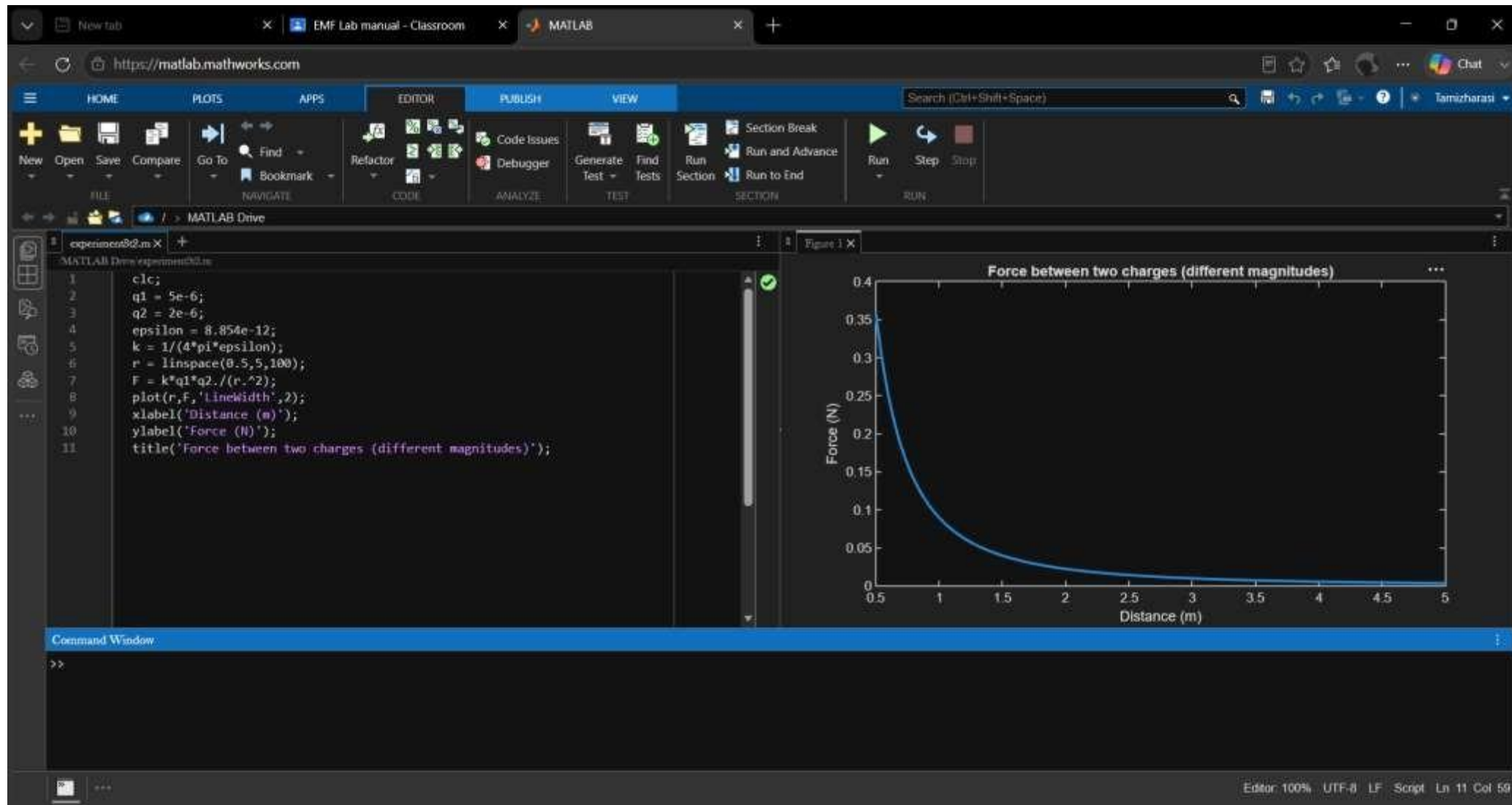


Exp. No: 3 Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance.

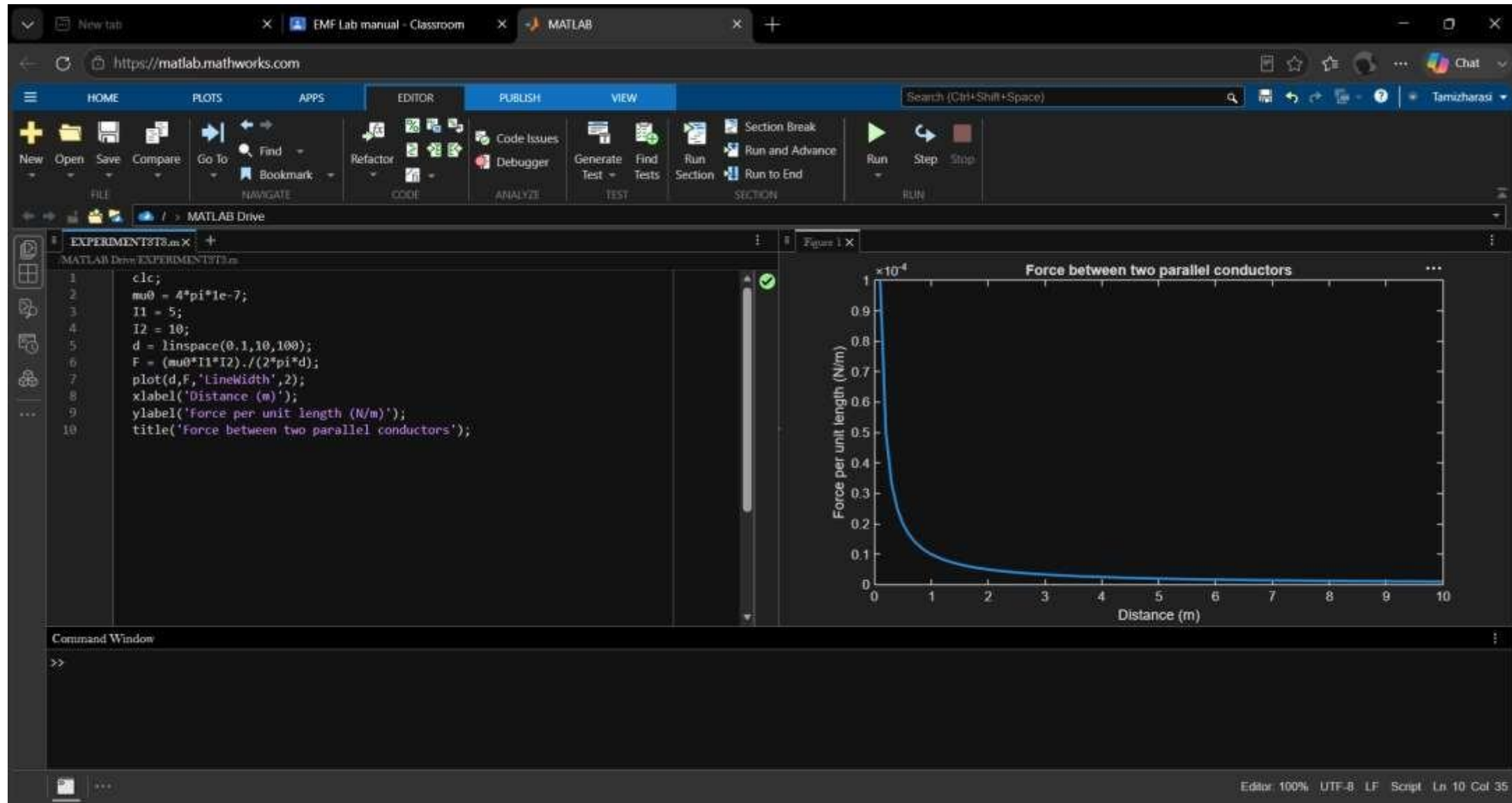
Test case (i) Determination and plotting of force between two electric charges with varying distance



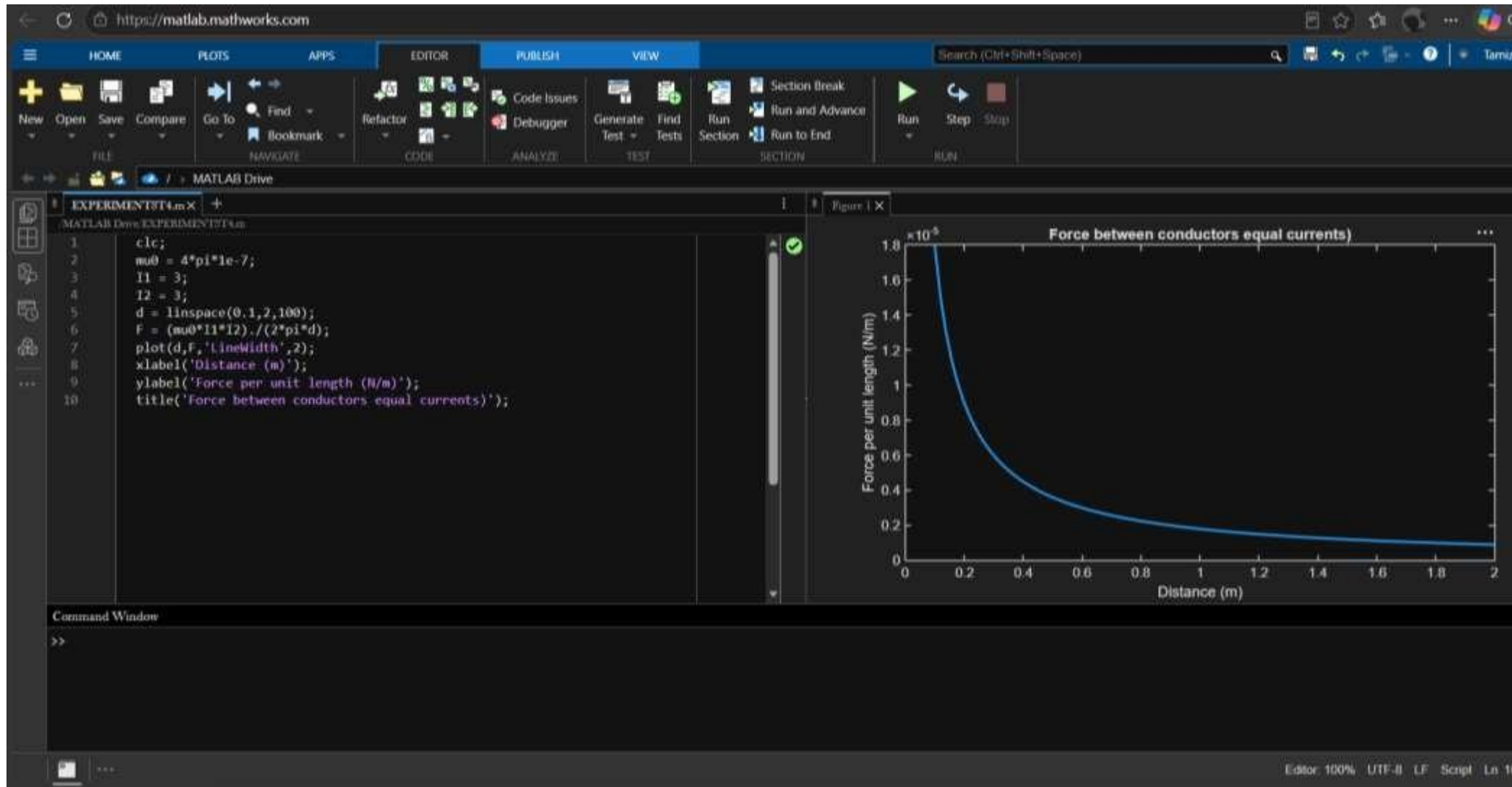
Test case (ii) Determination and plotting of force between two electric charges with different magnitudes



Test case (iii) Determination and plotting of force per unit length between two parallel

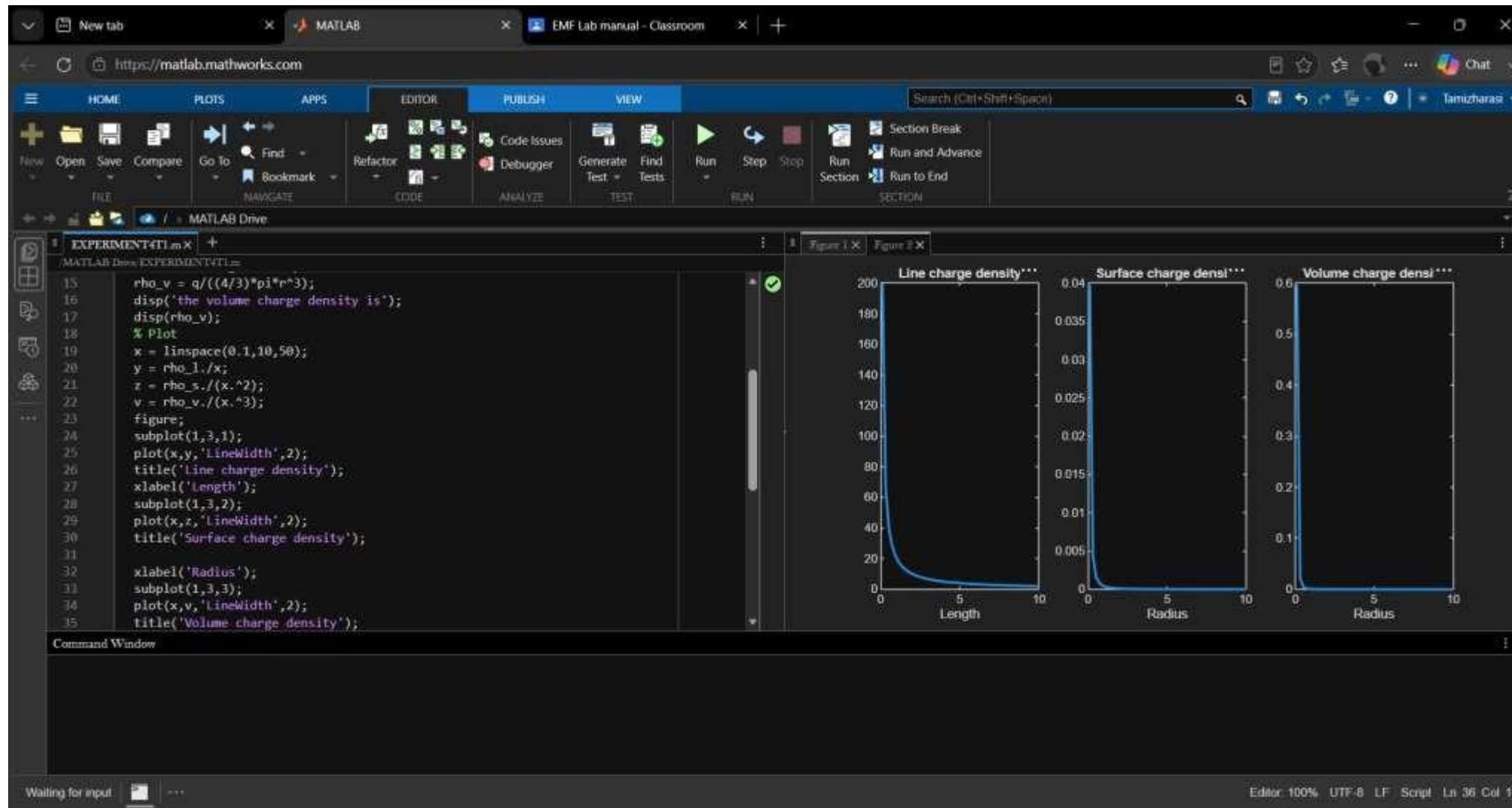


Test case (iv) Determination and plotting of force between two parallel conductors carrying equal currents

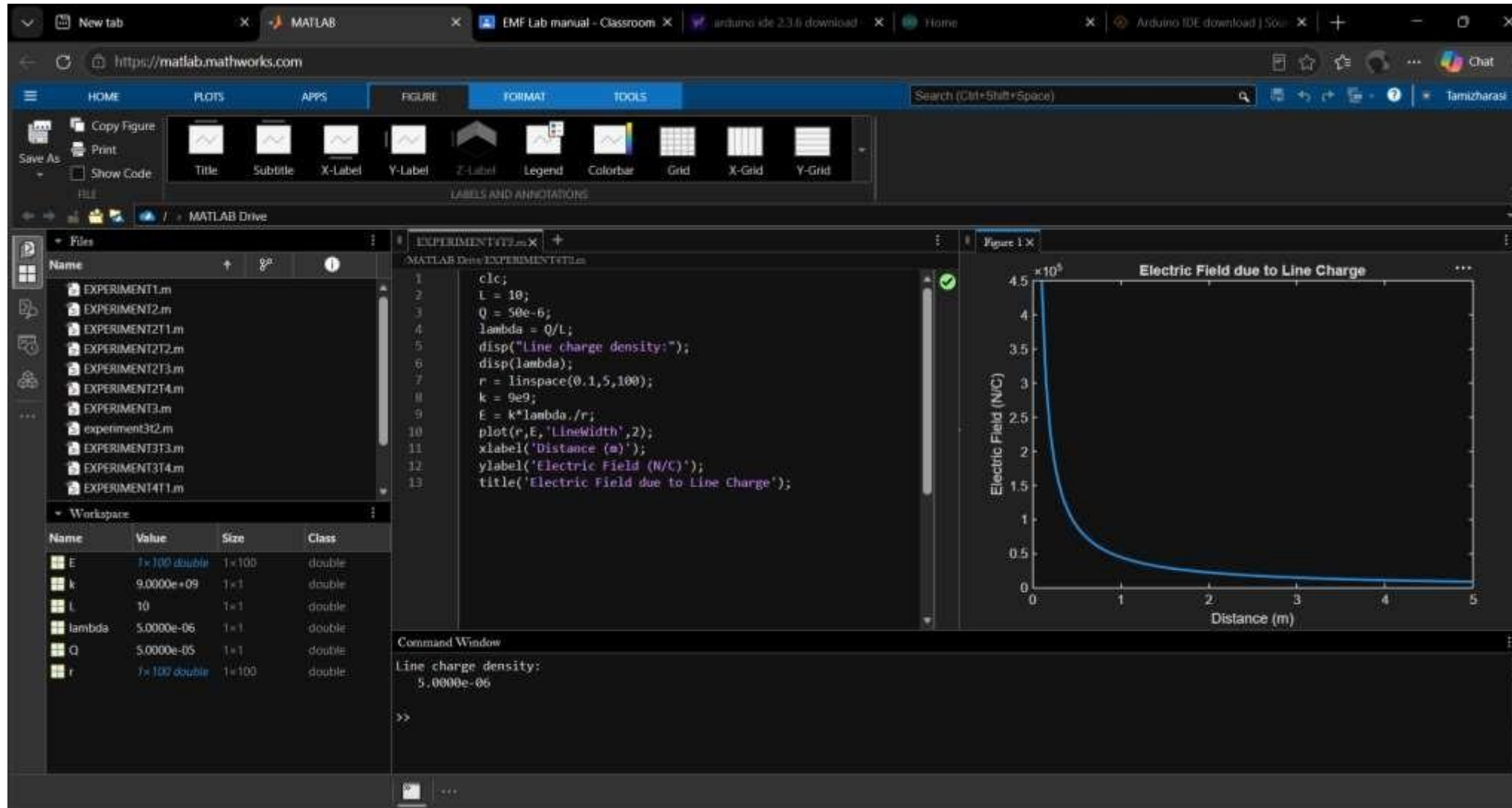


Exp.No:4

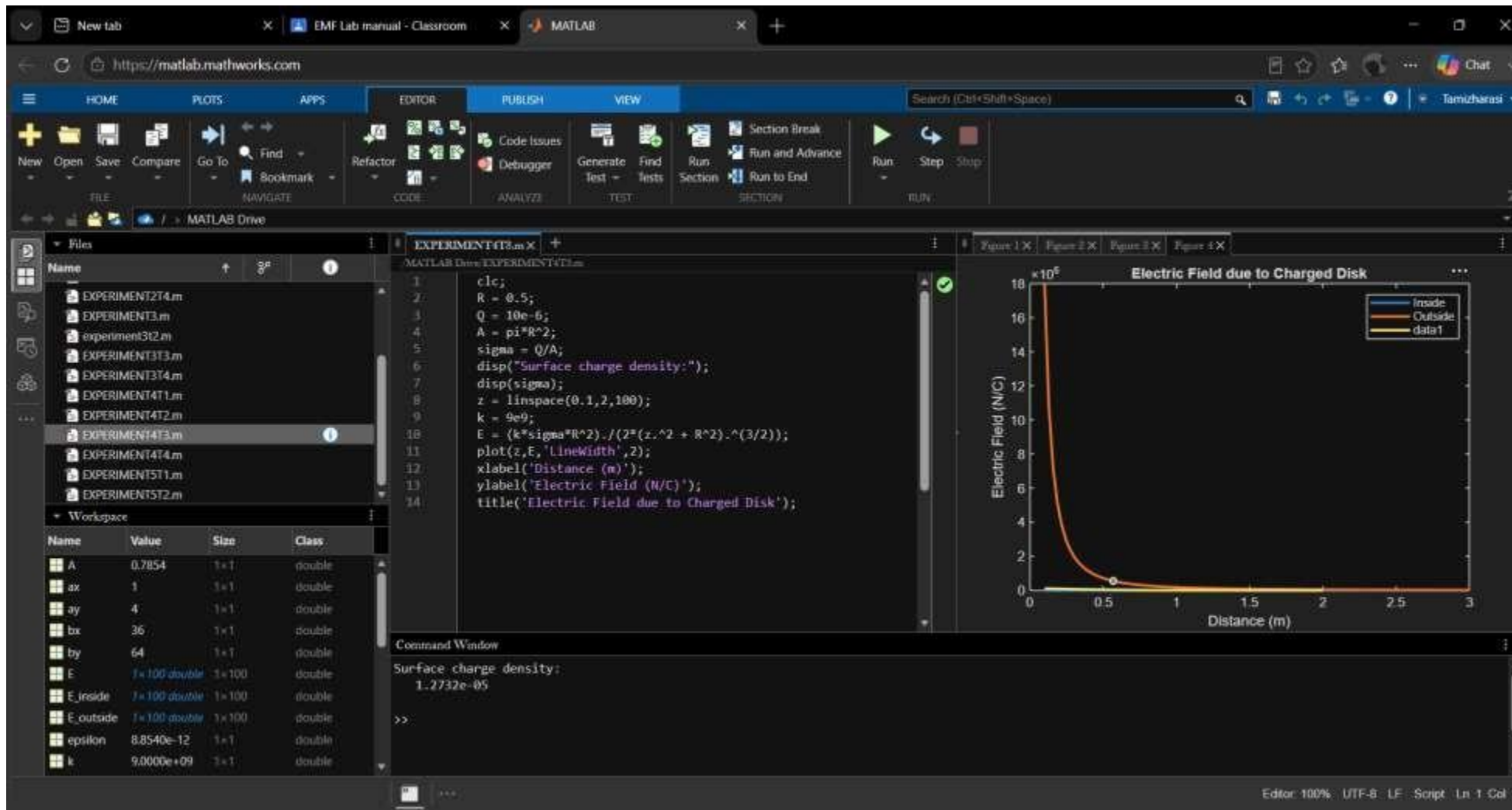
Test case (i) Determination of line, surface and volume charge density for given charge, length and radius



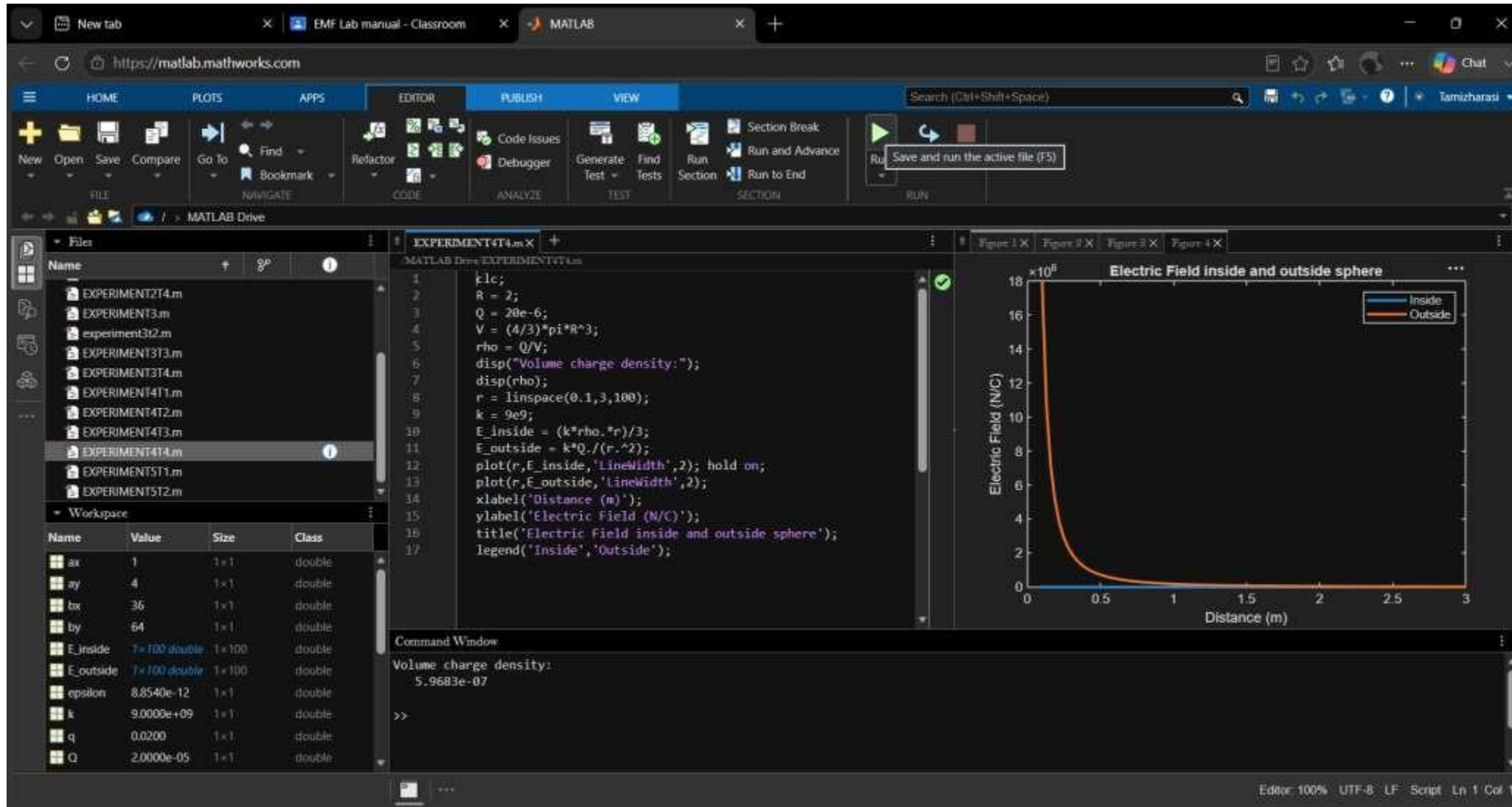
Test case (ii) A long wire of length $L = 10$ m carries a uniform charge of $Q = 50 \mu\text{C}$. Calculate the line charge density and plot electric field variation



Test case (iii) A uniformly charged disk of radius $R = 0.5$ m with total charge $Q = 10 \mu\text{C}$. Calculate surface charge density and electric field

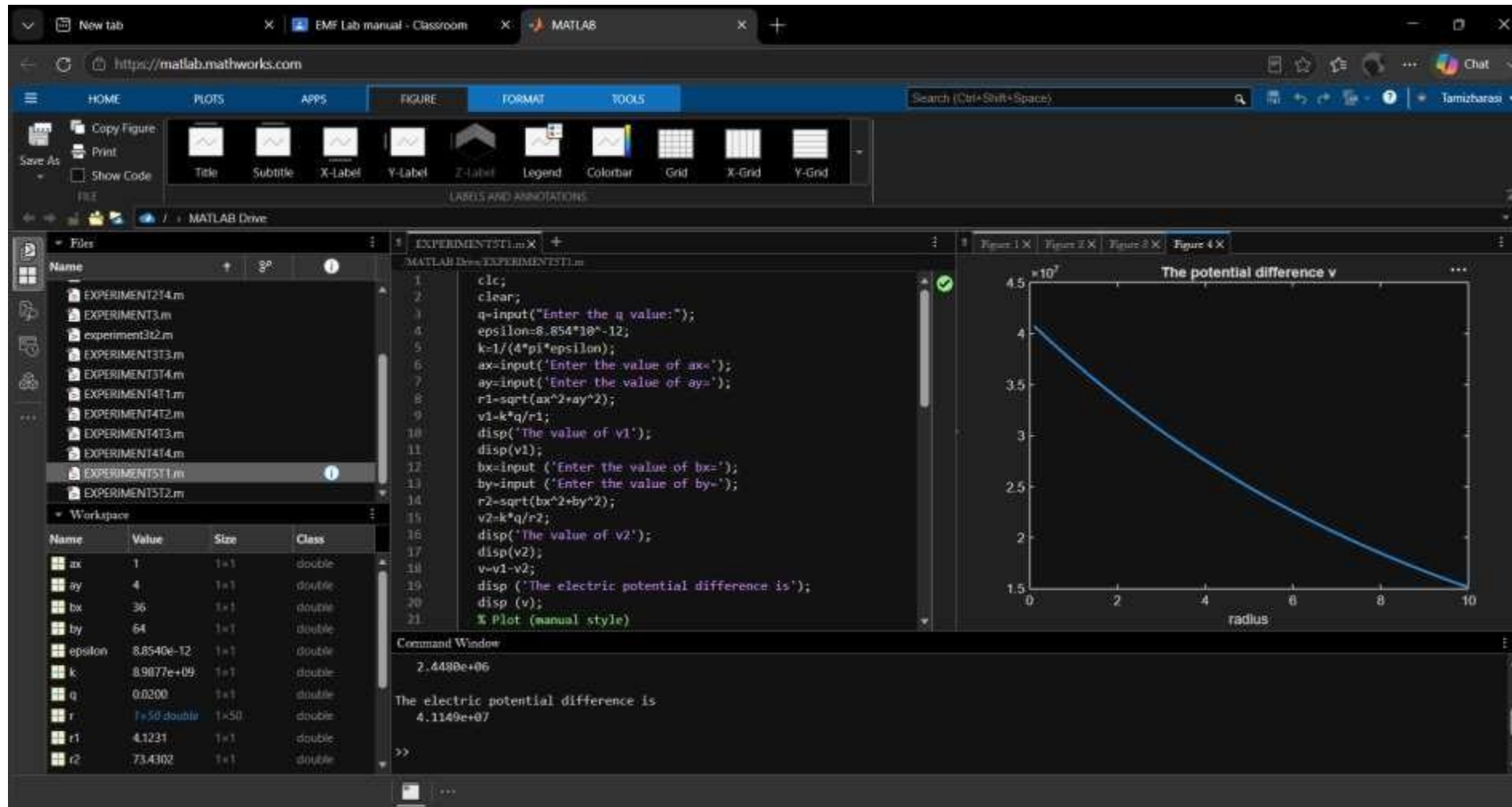


Test case (iv) A uniformly charged sphere of radius $R = 2$ m with total charge $Q = 20\mu\text{C}$. Calculate volume charge density and electric field inside and outside 3

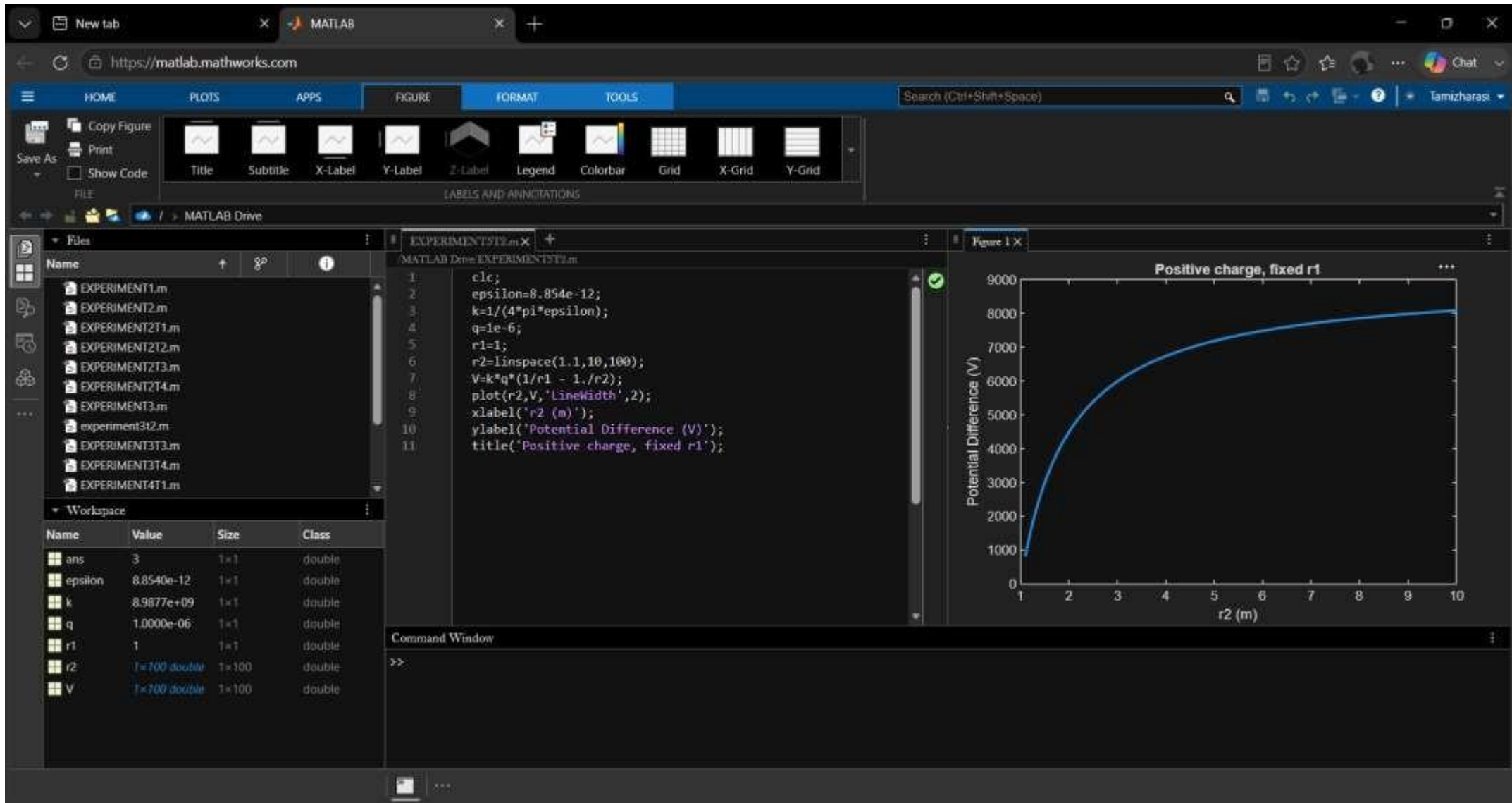


Exp. No: 5 Determination and plotting of Electric potential difference between any two points in free Space.

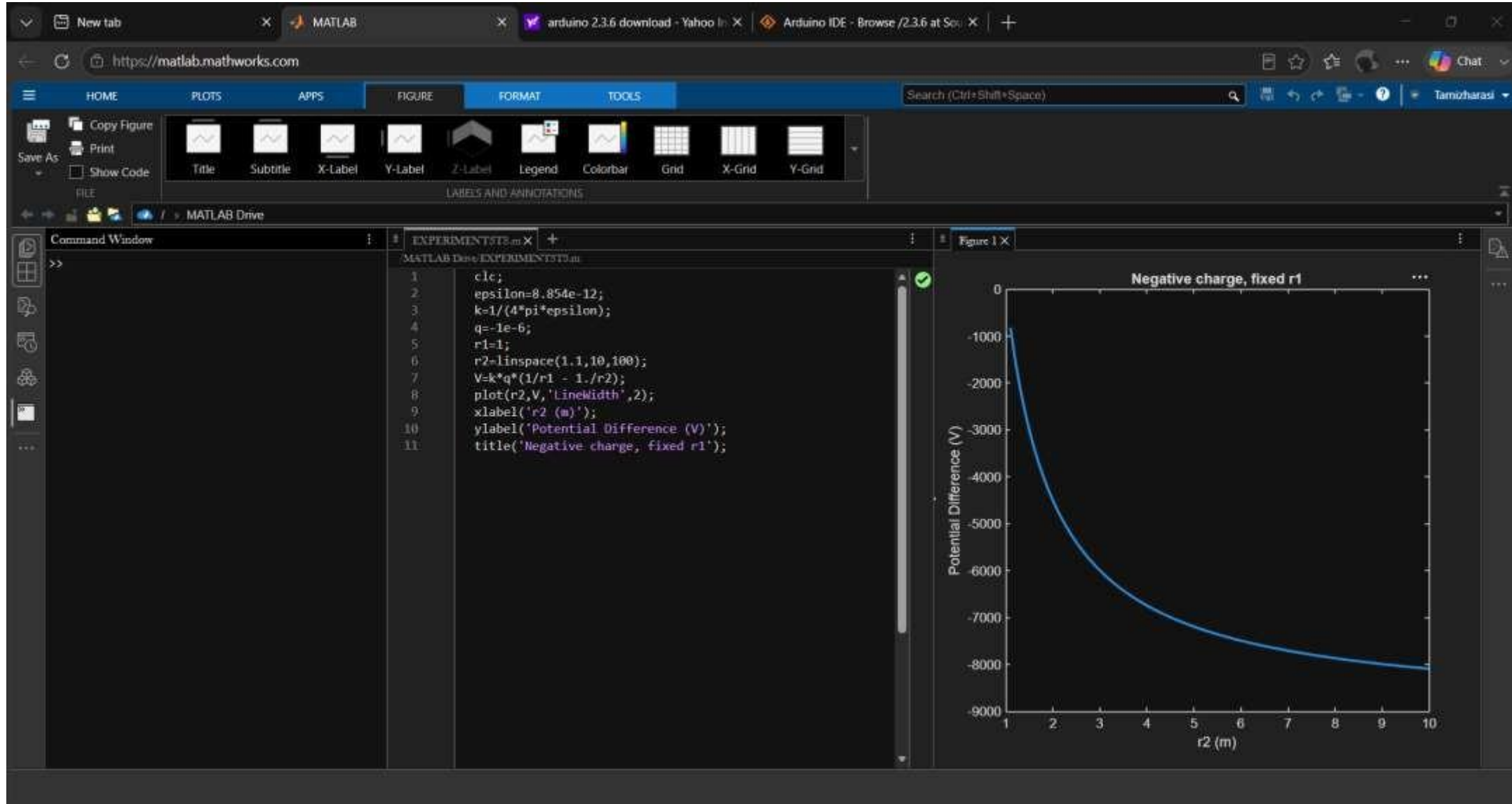
Test case 1:



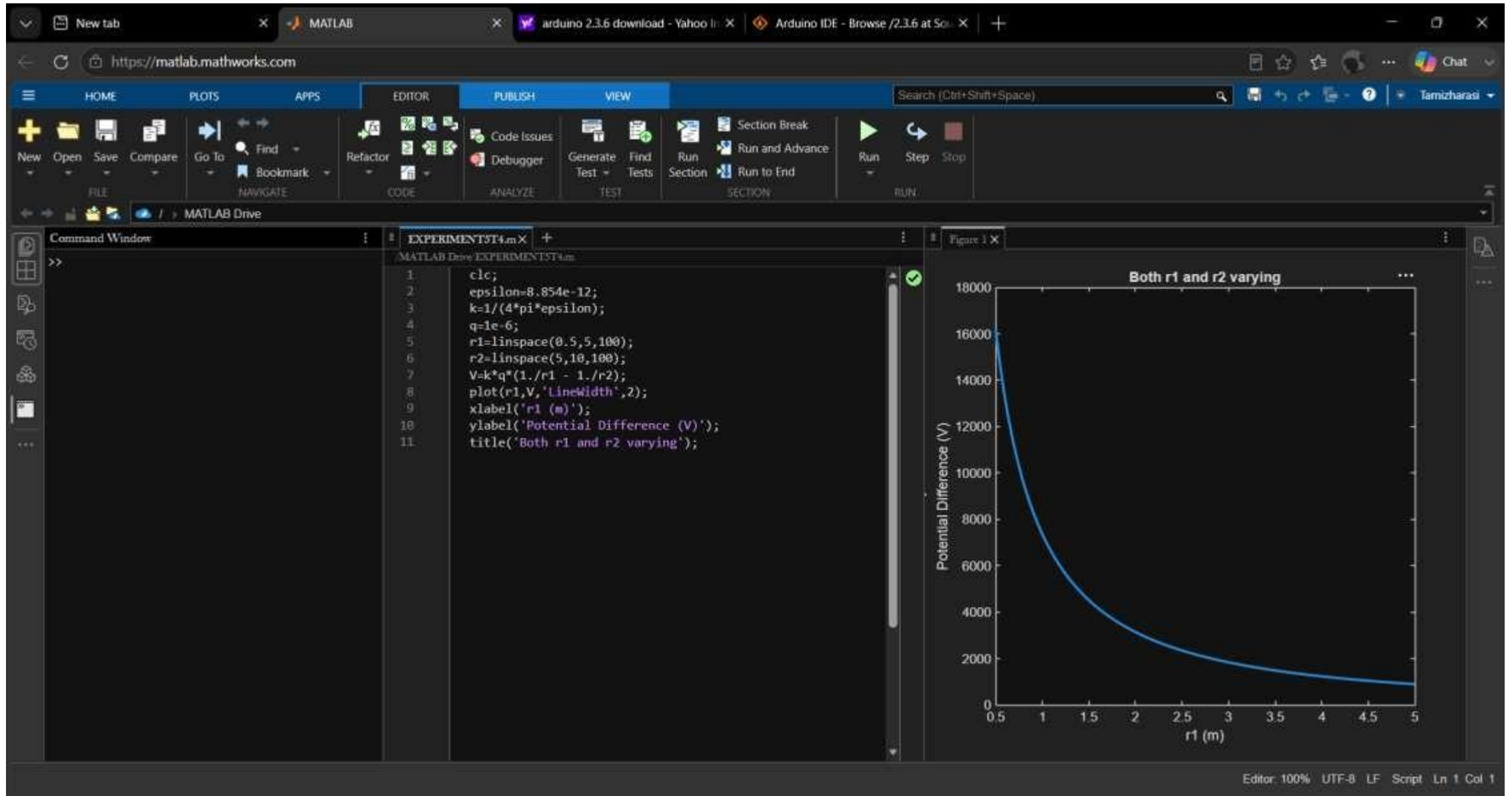
Test case (ii) A positive charge with fixed $r_1 = 1$ m and r_2 varying.



Test case (iii) A negative charge with fixed r_1 and varying r_2 :

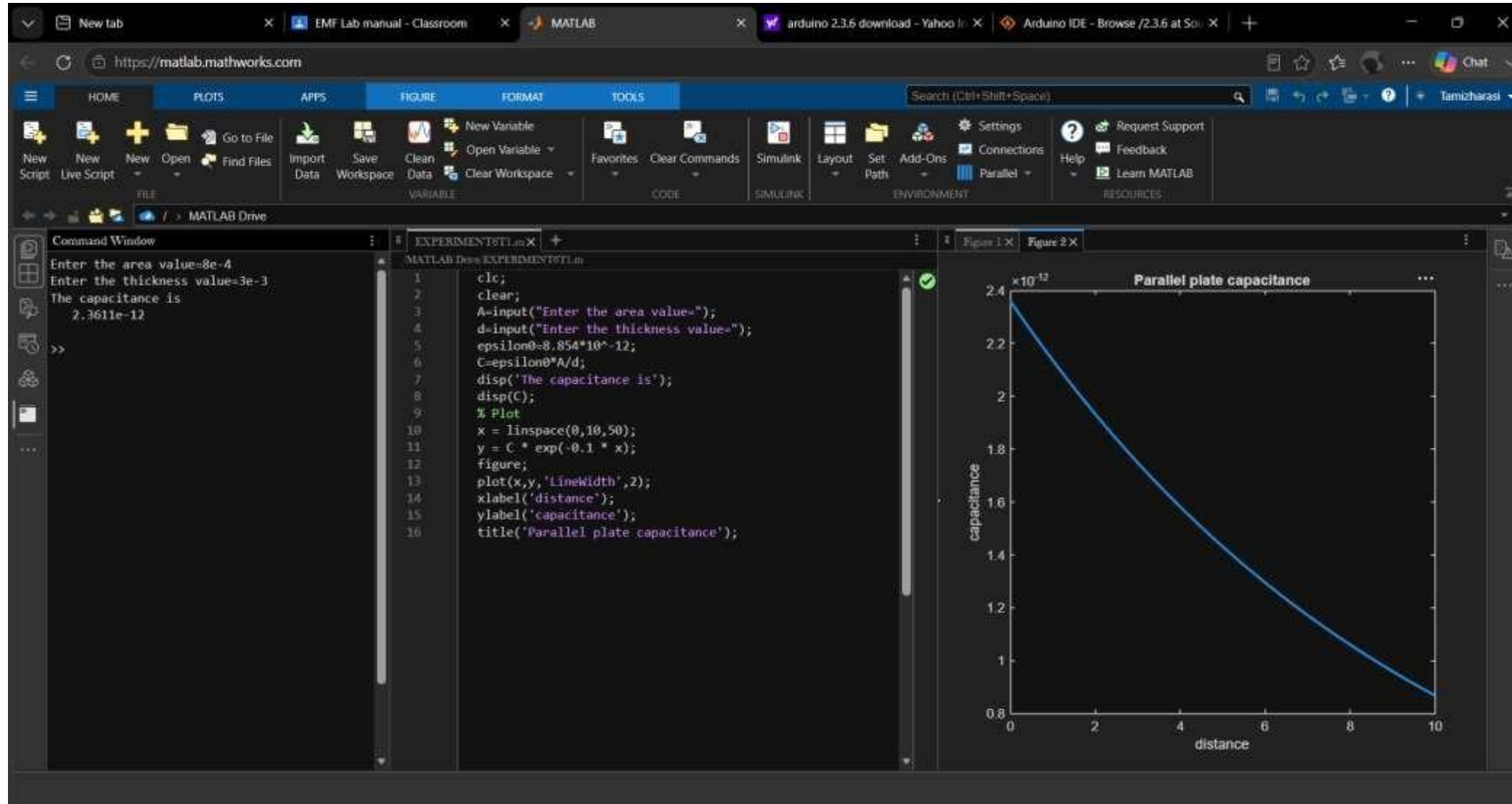


Test case (iv) Both r1 and r2 varying with positive charge 3:

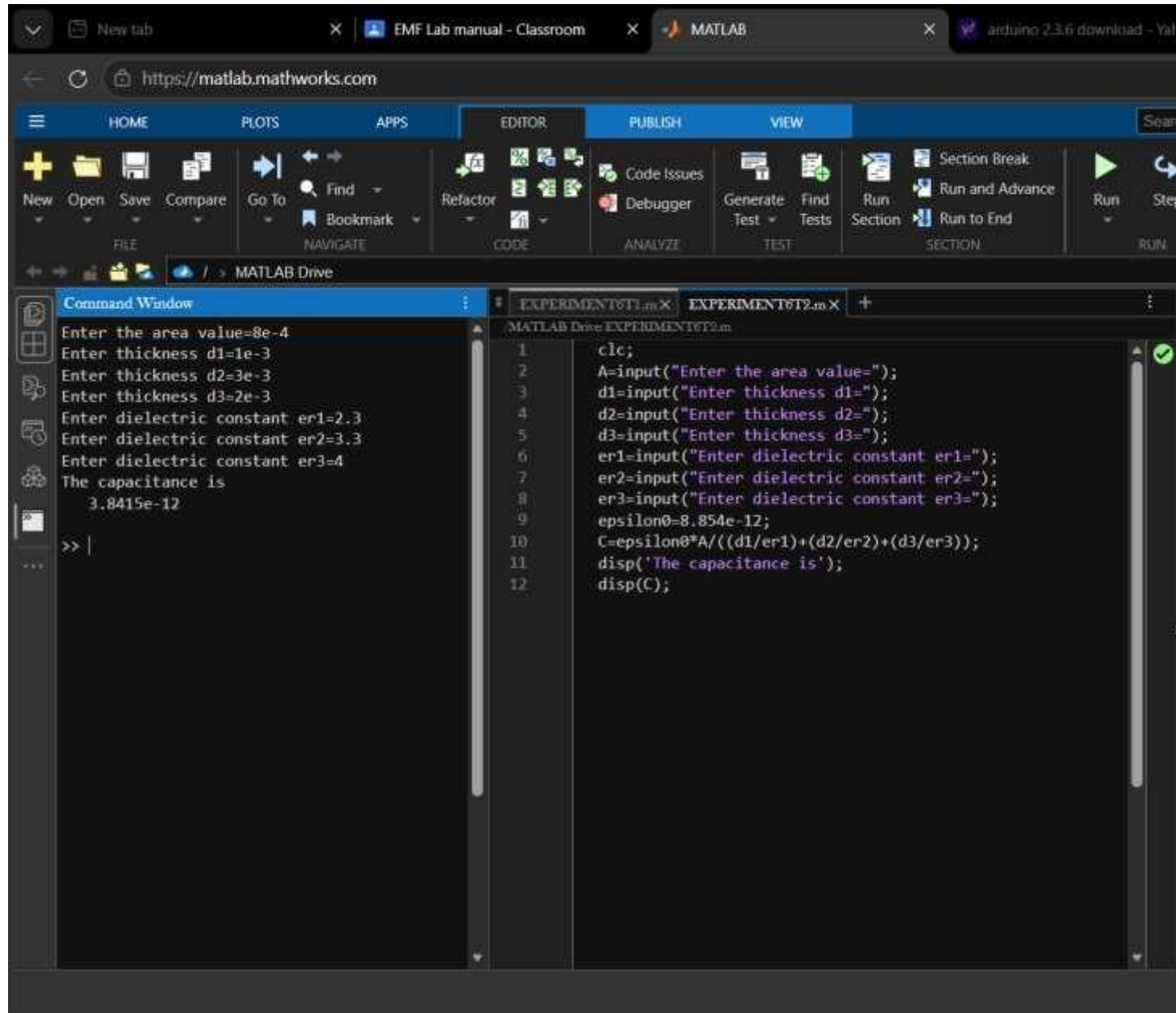


Exp no:6 Determination and plotting of Capacitance of parallel plate capacitor in Three different dielectric medium with specific area, dielectric thickness and various dielectric constants.

Test case (i) Determination and plotting of capacitance of parallel plate capacitor for given area and dielectric thickness



Test case (ii) Determination of capacitance of parallel plate capacitor with three different dielectric layers



The image shows a MATLAB web interface with a script for calculating the capacitance of a parallel plate capacitor with three dielectric layers. The Command Window displays the input values and the resulting capacitance.

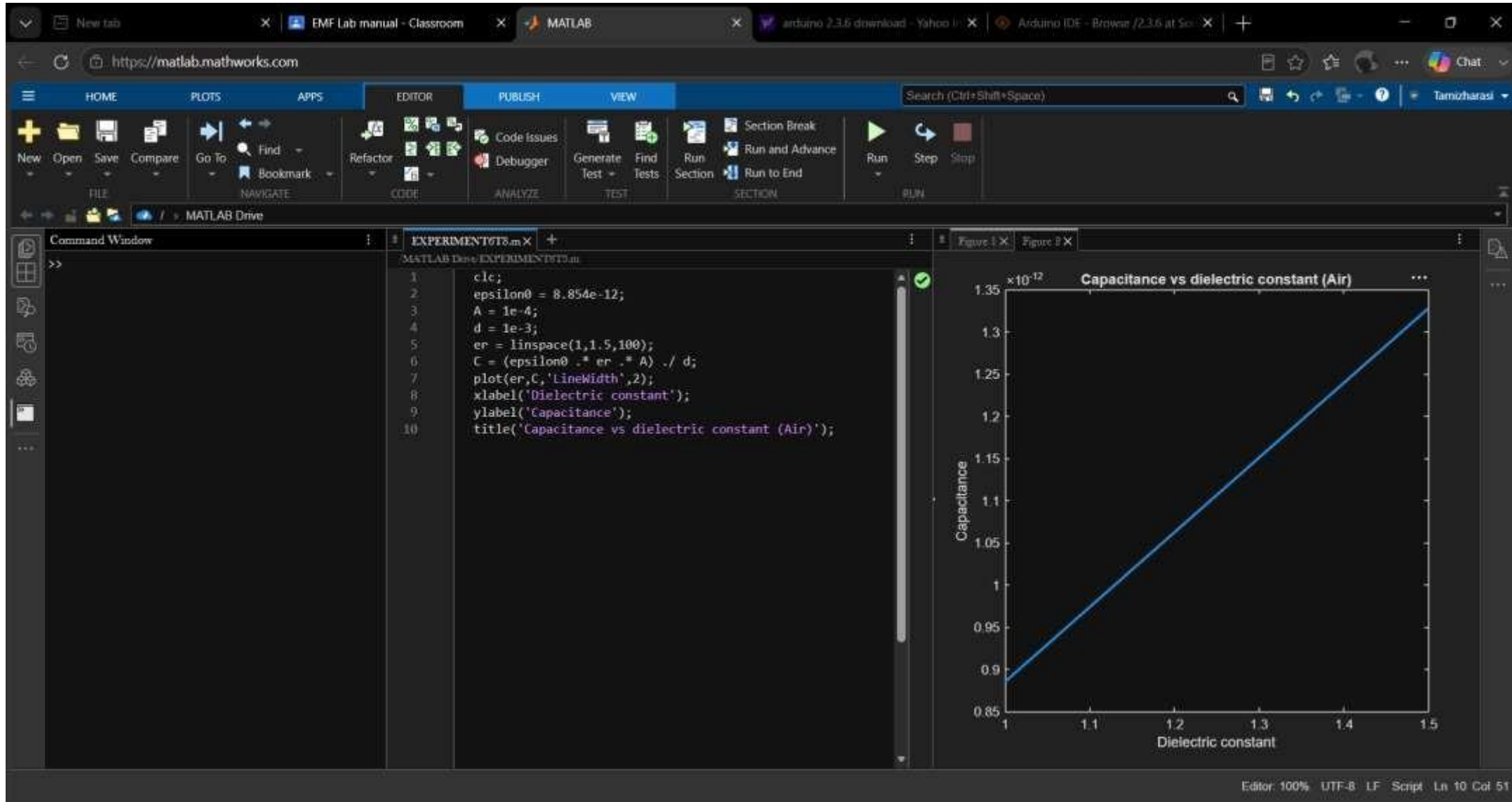
Command Window:

```
Enter the area value=8e-4  
Enter thickness d1=1e-3  
Enter thickness d2=3e-3  
Enter thickness d3=2e-3  
Enter dielectric constant er1=2.3  
Enter dielectric constant er2=3.3  
Enter dielectric constant er3=4  
The capacitance is  
3.8415e-12  
>> |
```

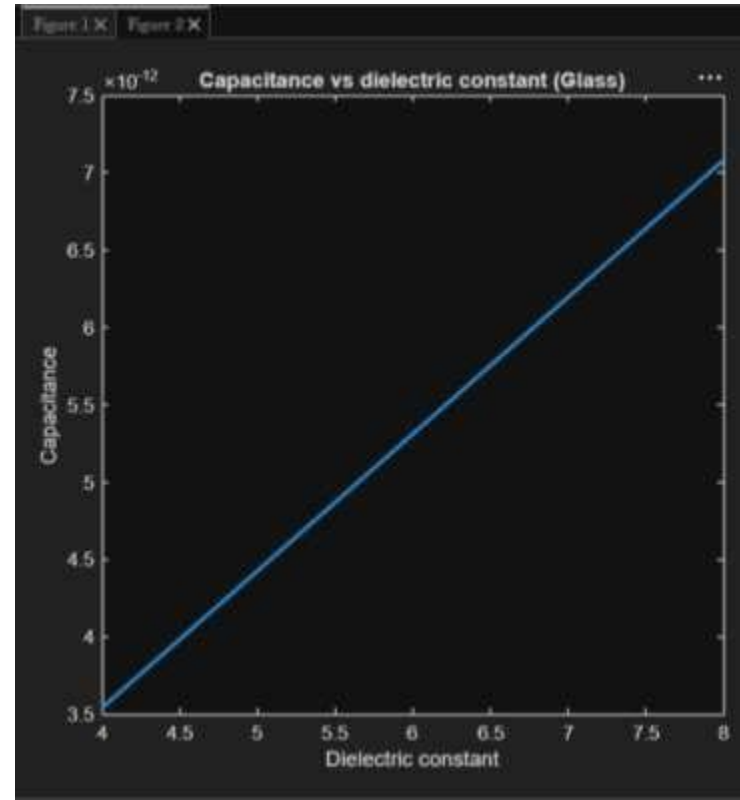
Script (EXPERIMENT6T1.m):

```
1 clc;  
2 A=input("Enter the area value=");  
3 d1=input("Enter thickness d1=");  
4 d2=input("Enter thickness d2=");  
5 d3=input("Enter thickness d3=");  
6 er1=input("Enter dielectric constant er1=");  
7 er2=input("Enter dielectric constant er2=");  
8 er3=input("Enter dielectric constant er3=");  
9 epsilon0=8.854e-12;  
10 C=epsilon0*A/((d1/er1)+(d2/er2)+(d3/er3));  
11 disp('The capacitance is');  
12 disp(C);
```

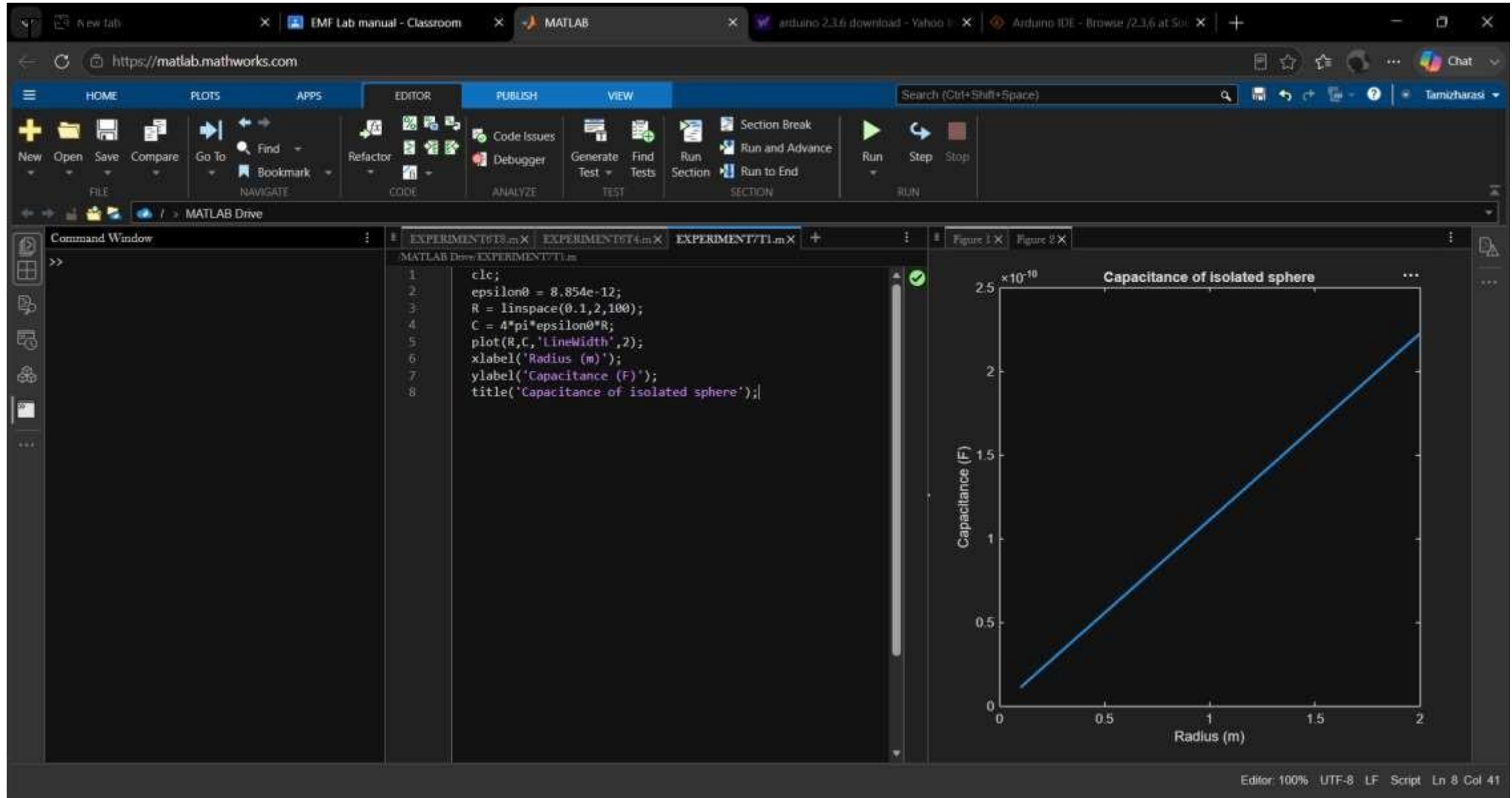

Test case (iii) Determination and plotting of capacitance variation with dielectric constant (Air)



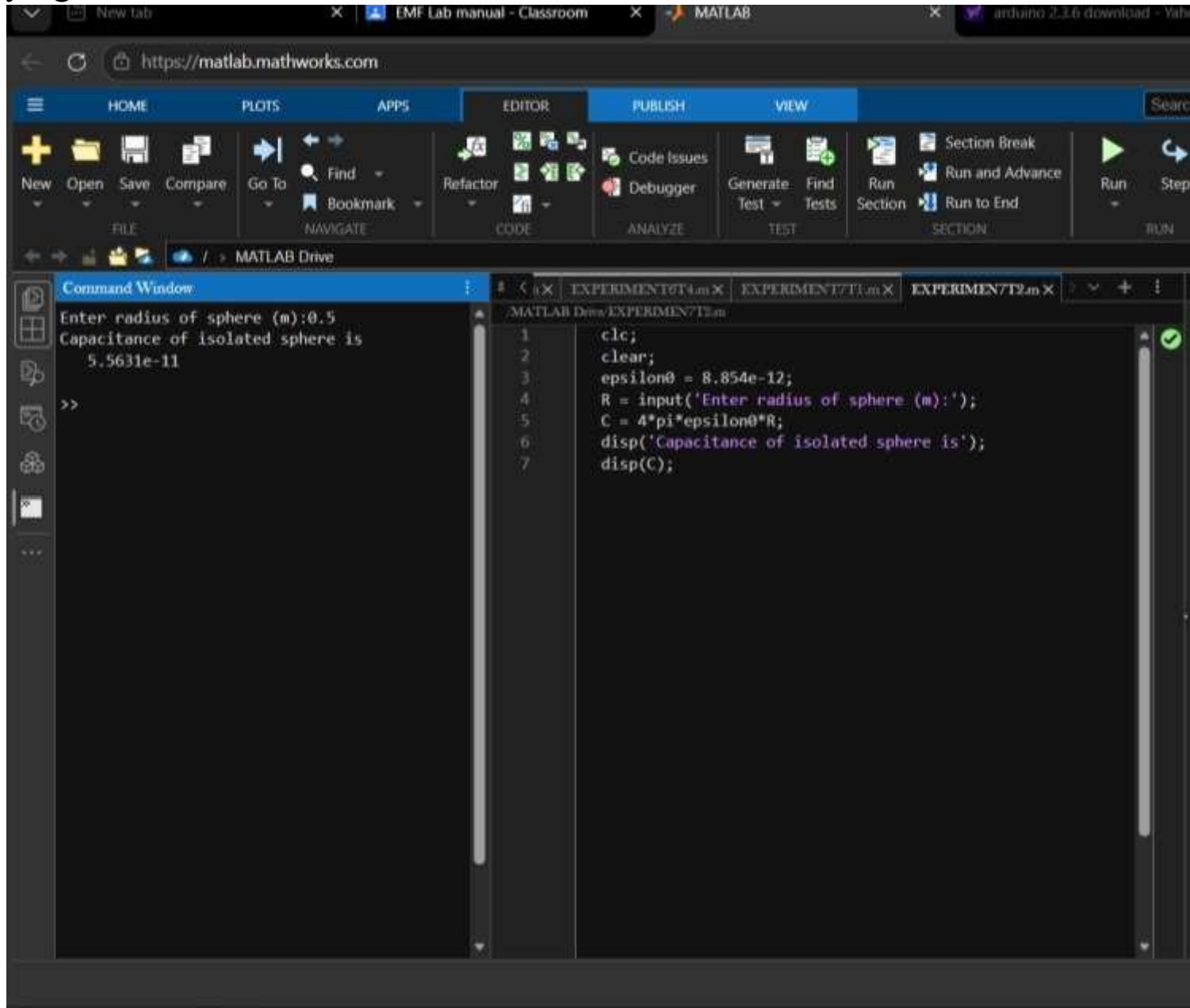
Test case (iv) Determination and plotting of capacitance variation with Dielectric constant (Glass)



Exp:7- Determination and plotting of Capacitance of isolated sphere and co-axial cable with different dimensions and dielectric constant.



Test case (ii) Determination and plotting of capacitance of isolated sphere with varying radius



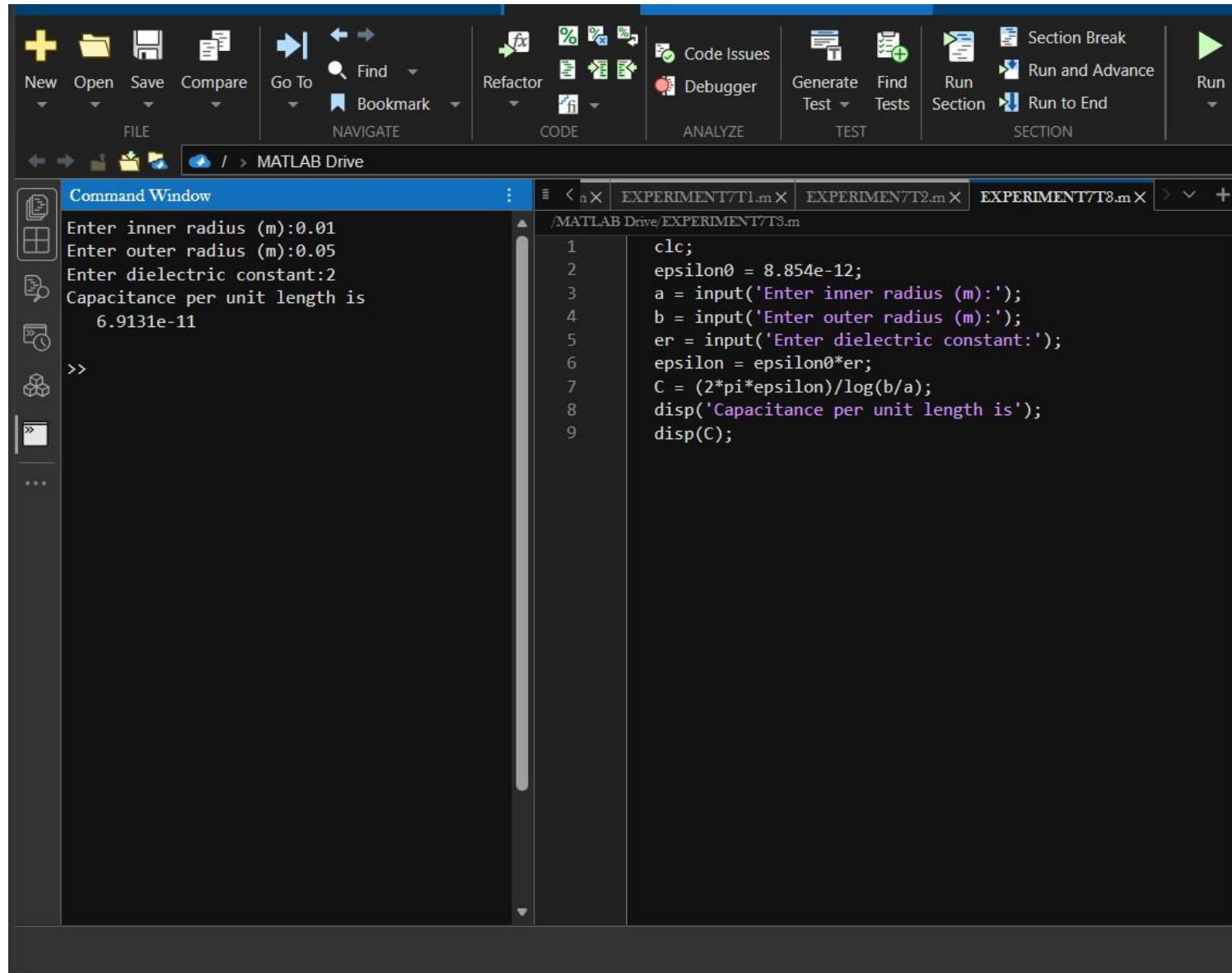
The image shows the MATLAB web interface with the following components:

- Browser Tabs:** New tab, EMF Lab manual - Classroom, MATLAB, arduino 2.1.6 download - Yahoo.
- Address Bar:** <https://matlab.mathworks.com>
- Navigation Bar:** HOME, PLOTS, APPS, EDITOR, PUBLISH, VIEW. A search bar is located on the right.
- Toolbars:**
 - FILE:** New, Open, Save, Compare.
 - NAVIGATE:** Go To, Find, Bookmark.
 - CODE:** Refactor.
 - ANALYZE:** Code Issues, Debugger.
 - TEST:** Generate Test, Find Tests.
 - SECTION:** Run Section, Run and Advance, Run to End.
 - RUN:** Run, Step.
- Command Window:**

```
Enter radius of sphere (m):0.5
Capacitance of isolated sphere is
5.5631e-11
>>
```
- Editor:** Shows the script `EXPERIMENT7T2.m` with the following code:

```
1 clc;
2 clear;
3 epsilon0 = 8.854e-12;
4 R = input('Enter radius of sphere (m):');
5 C = 4*pi*epsilon0*R;
6 disp('Capacitance of isolated sphere is');
7 disp(C);
```

Test case (iii) Determination of capacitance per unit length of co-axial cable



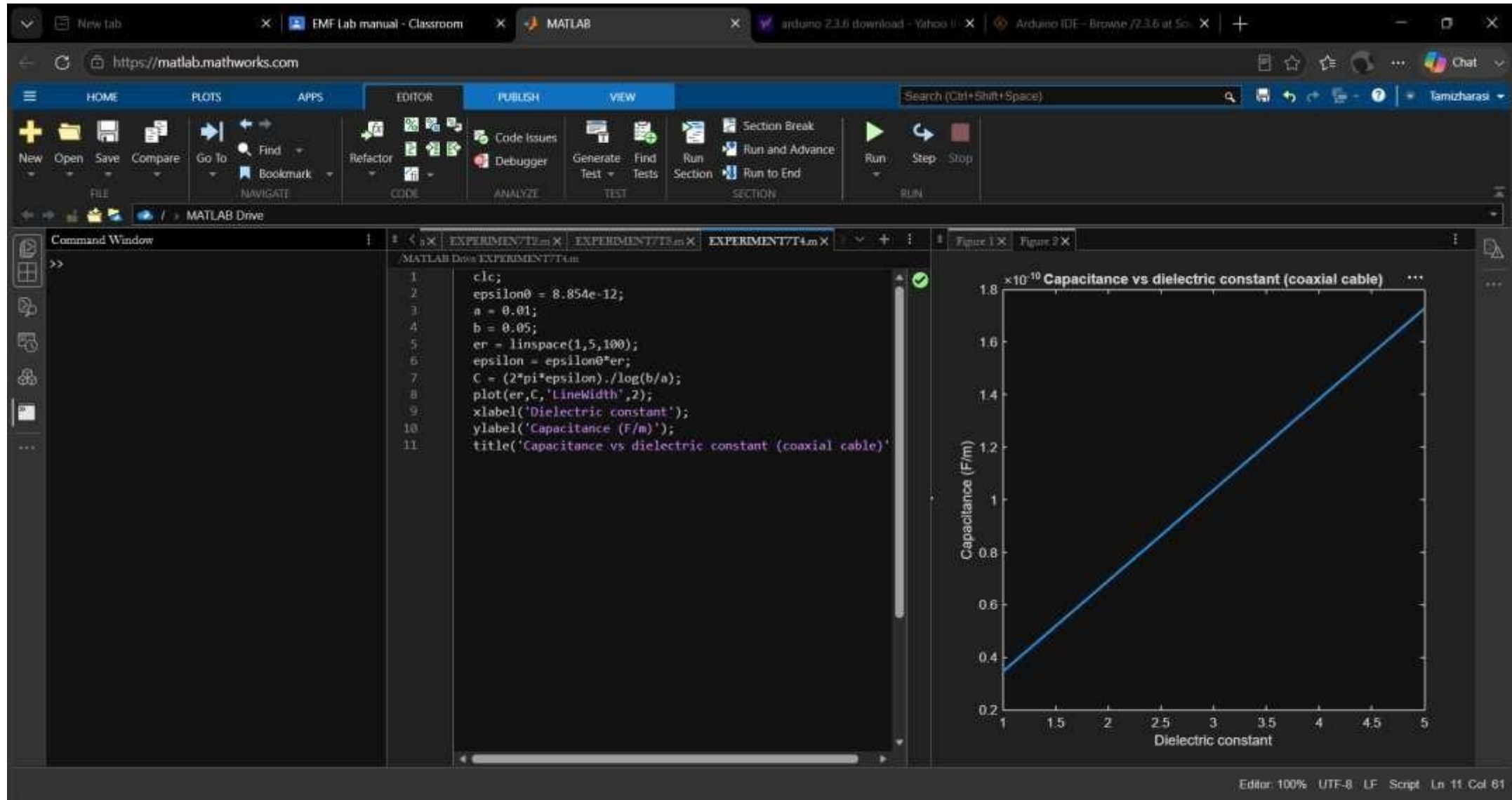
The image shows the MATLAB IDE interface. The top toolbar includes icons for File (New, Open, Save, Compare), Navigate (Go To, Find, Bookmark), Code (Refactor), Analyze (Code Issues, Debugger), Test (Generate Test, Find Tests), and Section (Section Break, Run, Run and Advance, Run to End). The Command Window on the left displays the following text:

```
Enter inner radius (m):0.01
Enter outer radius (m):0.05
Enter dielectric constant:2
Capacitance per unit length is
    6.9131e-11
>>
```

The Editor window on the right shows the script `EXPERIMENT7/T3.m` with the following code:

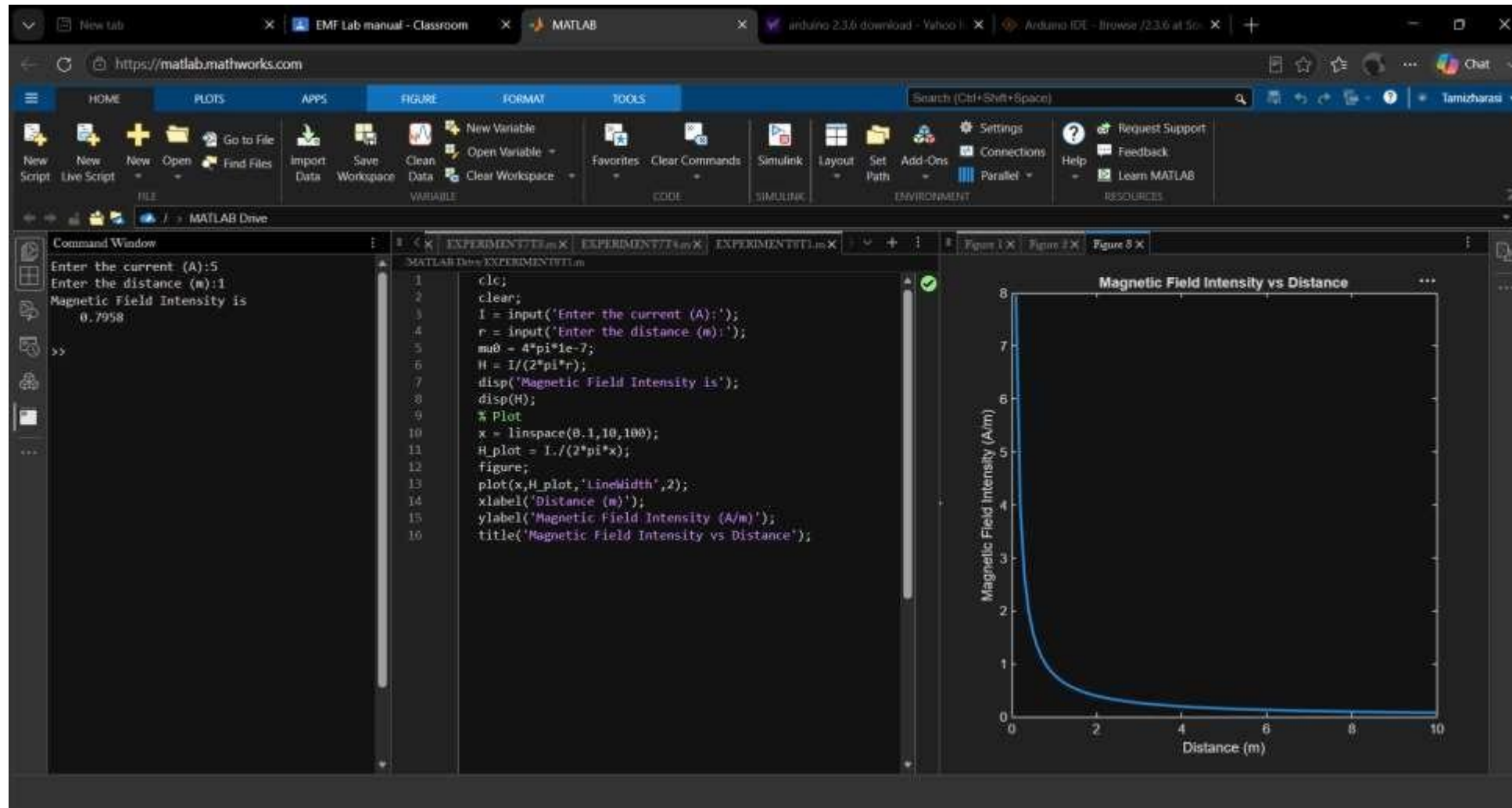
```
1 clc;
2 epsilon0 = 8.854e-12;
3 a = input('Enter inner radius (m):');
4 b = input('Enter outer radius (m):');
5 er = input('Enter dielectric constant:');
6 epsilon = epsilon0*er;
7 C = (2*pi*epsilon)/log(b/a);
8 disp('Capacitance per unit length is');
9 disp(C);
```


Test case (iv) Determination and plotting of capacitance variation with dielectric constant in co-axial cable

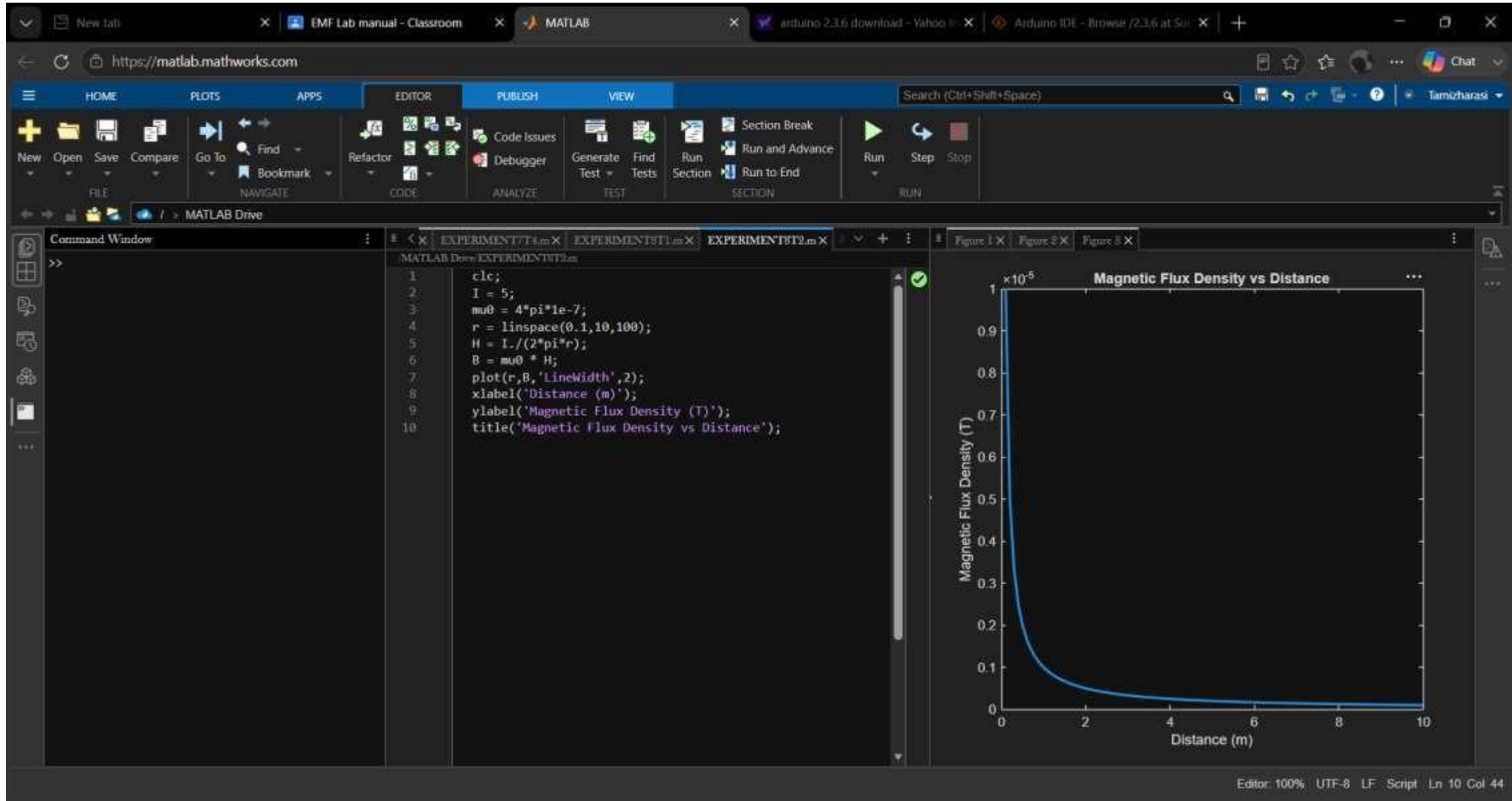


Exp no:8 - Determination and plotting of Magnetic Field Intensity, Intrinsic Impedance and Phase Velocity in different medium.

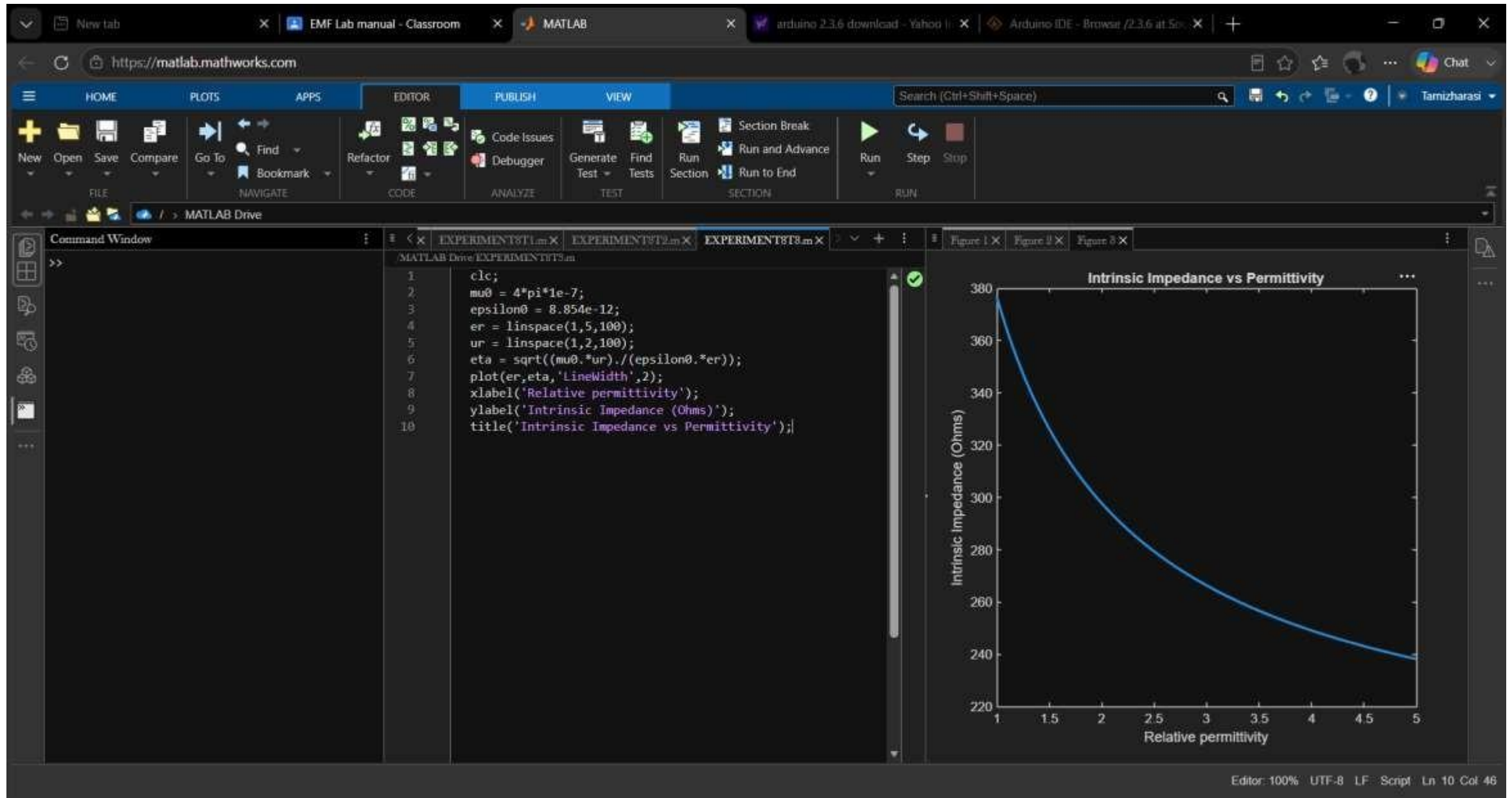
Test case (i) Determination and plotting of Magnetic Field Intensity due to infinite long conductor at a distance r



Test case (ii) Determination and plotting of Magnetic Flux Density in free space



Test case (iii) Determination and plotting of intrinsic impedance for different media



est case (iv) Determination and plotting of phase velocity in different medium

